LOAN RISK PREDICTION AND CREDIT RISK ANALYSIS

Data Scientist Intern Assessment — TapLend

Submitted by: Priya Rani
Course: B.Tech in Computer Science and Engineering
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1. Executive Summary

The objective of this project is to develop a machine learning-based loan risk prediction system that identifies applicants who may have a higher probability of loan default.

The analysis combines historical loan application information with loan outcome labels. Data preprocessing, exploratory data analysis, feature engineering, model training, and evaluation were performed using Python and Scikit-learn.

Two classification models, Logistic Regression and Random Forest, were evaluated. Random Forest achieved a bad-loan recall of approximately 72% and a ROC-AUC score of 0.710 in the customer-wise evaluation at the default classification threshold of 0.50.

To identify more potentially risky applicants, a threshold of 0.40 was also evaluated. At this threshold, bad-loan recall increased to approximately 94%, with a bad-loan precision of approximately 70%.

The final prediction analysis covered 130 applicant records. The model classified 110 applicants (84.62%) as high risk and 20 applicants (15.38%) as lower risk.

These results demonstrate the potential of machine learning for preliminary credit risk assessment. However, the model requires further validation before being used in real lending decisions.

2. Problem Statement

Lending organizations need to assess the repayment risk associated with loan applicants. Incorrectly approving high-risk applicants can result in financial losses, while incorrectly rejecting creditworthy applicants can reduce business opportunities.

The objective of this project is to use historical application data and loan outcome labels to identify applicant profiles associated with higher loan default risk.

The main objectives are:

- Analyze historical loan application and outcome data.
- Identify useful financial, demographic, and behavioral features.
- Develop machine learning models to predict bad-loan risk.
- Compare model performance using appropriate evaluation metrics.
- Generate risk probabilities and classify applicants into high-risk and lower-risk groups.
- Discuss limitations and recommend improvements for a practical lending system.

3. Dataset Description

The dataset was provided directly by **TapLend** as part of the data science assessment. It was used to analyze historical loan application data, study loan repayment outcomes, and develop machine learning models for loan risk prediction.

The dataset contains loan application information and corresponding loan outcome labels.

The application data includes information such as:

- Amount requested by the applicant.
- Age derived from birth date and application date.
- Monthly income and monthly rent.
- Bank account and direct deposit information.
- Credit-related scores, including FICO-based scores.
- Residence duration and bank account duration.
- Payment frequency and phone-related information.
- Intended use of the loan.

The outcome data contains loan identifiers and labels indicating whether a loan was classified as Good or Bad.

3.1 Data Integration

The application and outcome datasets were integrated using customer and loan identifier information.

The merged dataset contained **656 loan/application records representing 640 unique customers**. Sixteen customers had multiple records, making it important to account for repeated customers during model evaluation.

The target label was encoded as:

- **Bad Loan = 1**
- **Good Loan = 0**

This encoding makes the bad-loan class the positive class for risk-focused evaluation.

3.2 Target Distribution

The merged data contained:

- Bad-loan records: 354
- Good-loan records: 302

This distribution was examined before training to understand the relative representation of both outcome classes.

4. Technology Stack

The following tools and libraries were used:

- **Python:** Main programming language.
- **Jupyter Notebook:** Interactive development and documentation.
- **Pandas:** Data loading, cleaning, transformation, and analysis.
- **NumPy:** Numerical operations.
- **Matplotlib and Seaborn:** Data visualization.
- **Scikit-learn:** Preprocessing pipelines, model training, and evaluation.
- **Joblib:** Saving and loading the trained model.
- **JSON:** Storing model configuration, including the selected threshold.

5. Data Preprocessing and Feature Engineering

5.1 Data Cleaning

The preprocessing stage involved checking data types, missing values, identifiers, and duplicate customer records.

Date columns were converted into appropriate datetime formats. The validation results showed zero missing birth dates and zero missing application dates in the merged dataset used for the reported age analysis.

Identifier fields were excluded from the predictive feature set to prevent the model from learning arbitrary patterns from customer IDs.

5.2 Feature Engineering

Age was calculated from the difference between the application date and birth date.

Additional features included:

- Amount requested.
- Monthly income.
- Monthly rent.
- Credit-related scores.
- Direct deposit indicator.
- Residence duration.
- Bank account duration.
- Payment frequency.
- Loan purpose and phone-related categories.

Categorical variables were encoded using one-hot encoding. Numerical features were processed using median imputation and standard scaling, while categorical features were imputed using the most frequent category.

The preprocessing steps were incorporated into Scikit-learn pipelines to maintain consistent transformations during training and prediction.

5.3 Leakage Prevention

Potentially unavailable-at-application-time fields, such as approved loan terms or payment information recorded after the lending decision, must be reviewed before inclusion in a predictive model.

Only information that would legitimately be available when assessing a new applicant should be used for a deployable model.

6. Exploratory Data Analysis

Exploratory data analysis was used to understand the dataset, identify patterns, and assess the suitability of candidate features.

The analysis included:

- Inspecting data distributions and missing values.
- Examining age and income distributions.
- Reviewing credit-related scores.
- Checking repeated customer records.
- Understanding the target class distribution.
- Reviewing feature importance from the Random Forest model.

The feature importance analysis identified retail FICO score, age, bank-card FICO score, L2C score, telecom FICO score, money-related FICO score, monthly income, and monthly rent among the more influential features in the fitted Random Forest model.

Feature importance indicates how the model uses variables for prediction; it does not establish that these variables cause loan defaults.

7. Model Development

Two classification algorithms were trained and evaluated.

7.1 Logistic Regression

Logistic Regression was used as a baseline model. It estimates the probability of a binary outcome and provides a useful comparison against more complex models.

7.2 Random Forest

Random Forest combines predictions from multiple decision trees. It can model nonlinear relationships and interactions between features.

The implementation used 300 trees, a minimum leaf size of 3, and balanced class weights.

7.3 Evaluation Strategy

A customer-wise train-test split was used to reduce the risk of the same customer appearing in both training and testing sets.

The evaluation used 526 training records and 130 testing records, with no overlapping customers. The test set contained 82 bad-loan records and 48 good-loan records.

The following metrics were considered:

- **Precision:** The proportion of applicants predicted as bad loans who were actually bad loans.
- **Recall:** The proportion of actual bad loans correctly identified by the model.
- **F1-score:** The harmonic mean of precision and recall.
- **ROC-AUC:** The model's ability to rank bad-loan cases above good-loan cases across thresholds.
- **PR-AUC:** A precision-recall-based measure that is useful for evaluating positive-class detection.

8. Model Results and Comparison

The customer-wise evaluation at the default threshold of 0.50 produced the following results:

Metric	Logistic Regression	Random Forest
Accuracy	65.4%	64.6%
Bad-loan precision	76.1%	72.0%
Bad-loan recall	65.9%	72.0%
Bad-loan F1-score	70.6%	72.0%
ROC-AUC	0.694	0.710
PR-AUC	0.754	0.798

Random Forest achieved higher bad-loan recall, F1-score, ROC-AUC, and PR-AUC than Logistic Regression in this evaluation. Logistic Regression achieved slightly higher accuracy and bad-loan precision.

Because missing an actual bad loan can be costly for a lender, Random Forest was selected for further threshold analysis. The choice remains dependent on the organization's risk tolerance and the cost of incorrect decisions.

9. Classification Threshold Analysis

A classification threshold converts predicted probabilities into discrete risk classifications.

The default threshold of 0.50 was compared with alternative thresholds. The customer-wise test results showed the following behavior:

Threshold	Bad-loan precision	Bad-loan recall
0.30	66.1%	97.6%
0.40	70.0%	93.9%

0.50	72.0%	72.0%
0.60	81.8%	32.9%
0.70	100.0%	4.9%

At a threshold of 0.40, the model detected a greater proportion of actual bad loans than at 0.50, but it also classified more good loans as risky.

Therefore, a lower threshold may be useful when avoiding missed defaults is a priority, while a higher threshold may be appropriate when the lender wants to reduce false alarms. The threshold should be selected using validation data and a clearly defined business objective.

10. Final Applicant Risk Analysis

The selected Random Forest model and threshold of 0.40 were applied to 130 applicant records.

The resulting distribution was:

Risk Category	**Applicants**	**Percentage**
High Risk	110	84.62%
Lower Risk	20	15.38%
Total	**130**	**100%**

The mean predicted bad-loan probability was 0.5188, or approximately 51.88%.

The minimum predicted probability was 0.1916 (19.16%), while the maximum was 0.7684 (76.84%).

These figures represent model-estimated probabilities, not confirmed default outcomes. The large high-risk group may reflect the selected threshold, the applicant population, the features, or the model's calibration. It should not be interpreted as proof that 84.62% of these applicants will default.

11. Limitations

The current analysis has several limitations.

Limited observation period: The available historical data covers a relatively short period, which may not capture changing economic conditions or long-term repayment behavior.

Limited sample size: The dataset contains 656 records and 640 unique customers. Larger and more representative datasets would be needed for more reliable generalization.

Selection bias: The historical outcomes may represent loans that were actually issued rather than all applicants. The model may therefore not generalize to applicants who were previously rejected.

Repeated customer records: Some customers have multiple records. Customer-wise evaluation reduces direct overlap between training and testing customers, but repeated records and their temporal ordering still need careful handling.

Potential data leakage: Variables recorded after loan approval or during repayment could make offline performance appear better than real-world performance if used improperly.

Threshold sensitivity: The proportion of applicants classified as high risk changes substantially with the classification threshold.

Fairness and explainability: Credit decisions can affect people's access to financial services. Feature behavior, fairness across relevant groups, and explanations for individual predictions require further assessment.

Probability calibration: Predicted probabilities should be checked for calibration before being interpreted as reliable estimates of actual default likelihood.

12. Recommendations and Future Improvements

The following improvements could strengthen the model:

1. Collect a larger and more recent dataset covering a longer period.
2. Include verified repayment history, existing debt obligations, loan tenure, and consistent default definitions.
3. Add verified income and cash-flow information where legally permitted.
4. Conduct temporal validation using older records for training and newer records for testing.
5. Tune model parameters and classification thresholds using validation data rather than the final test set.
6. Evaluate probability calibration and monitor performance after deployment.
7. Perform fairness assessments and provide interpretable reasons for risk assessments.
8. Establish a monitoring process to detect changes in applicant characteristics and model performance.
9. Compare model performance against simple business rules and a baseline model.
10. Use predictions as decision support, with appropriate human oversight and applicable lending regulations.

13. Conclusion

This project demonstrates an end-to-end machine learning workflow for preliminary loan risk assessment, from data integration and feature engineering to model evaluation and applicant-level predictions.

Random Forest performed better than Logistic Regression on several risk-focused metrics in the customer-wise evaluation. A threshold of 0.40 increased bad-loan recall to approximately 94%, with precision of approximately 70%, illustrating the trade-off between identifying risky applicants and incorrectly flagging creditworthy applicants.

The final predictions classified 110 of 130 applicants as high risk and 20 as lower risk. These classifications are preliminary model outputs and should not be treated as confirmed repayment outcomes or used as the sole basis for lending decisions.

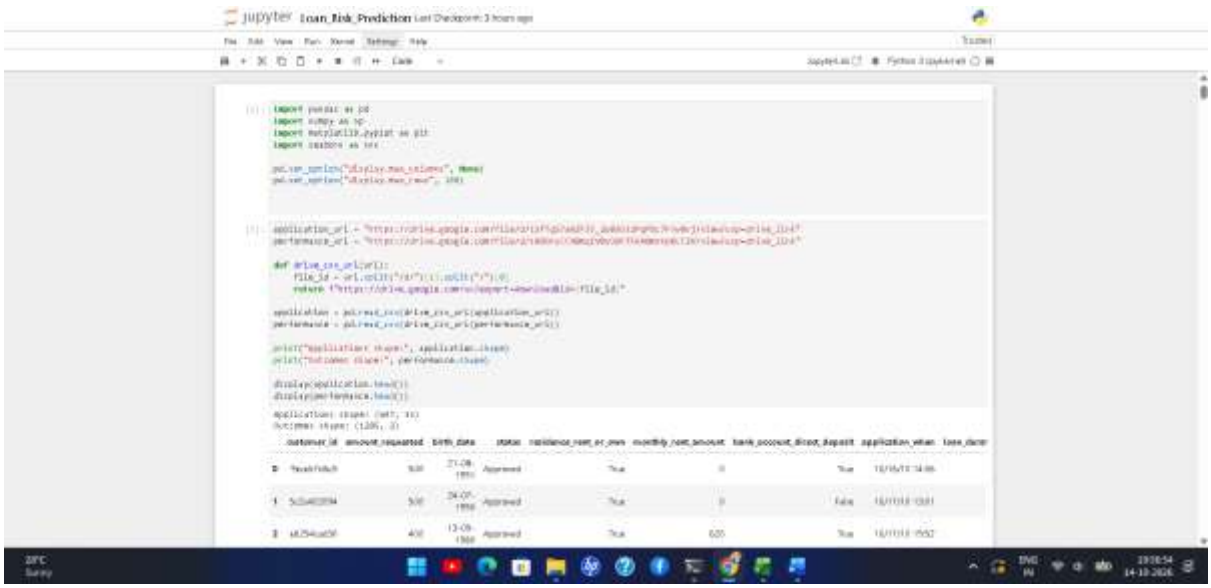
Further validation, more representative data, leakage checks, calibration, fairness analysis, and ongoing monitoring would be required before real-world deployment.

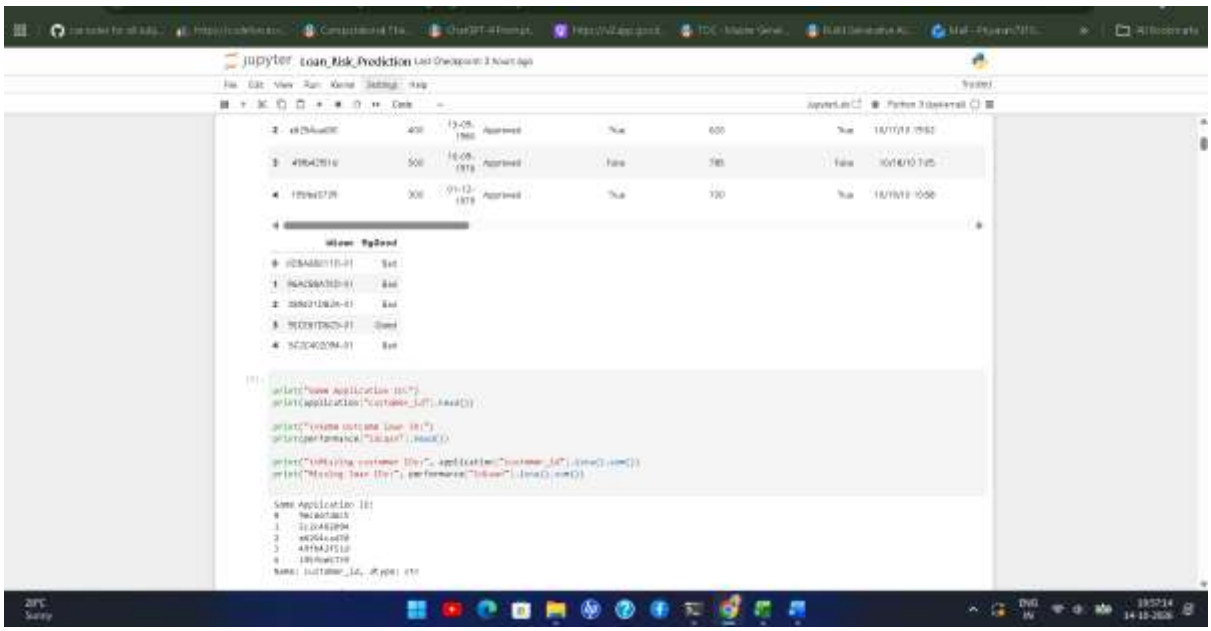
14. Deliverables

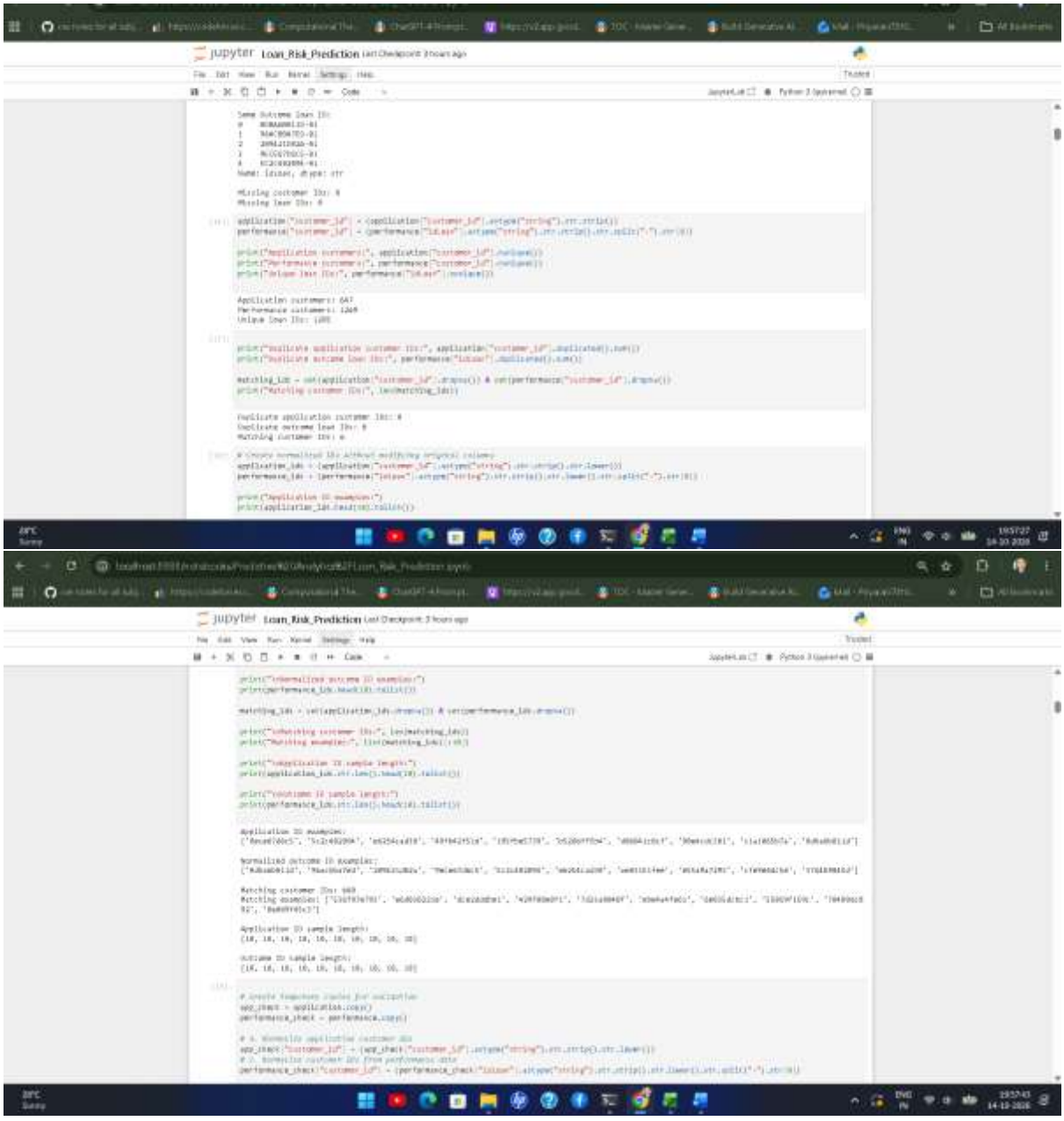
The project deliverables are:

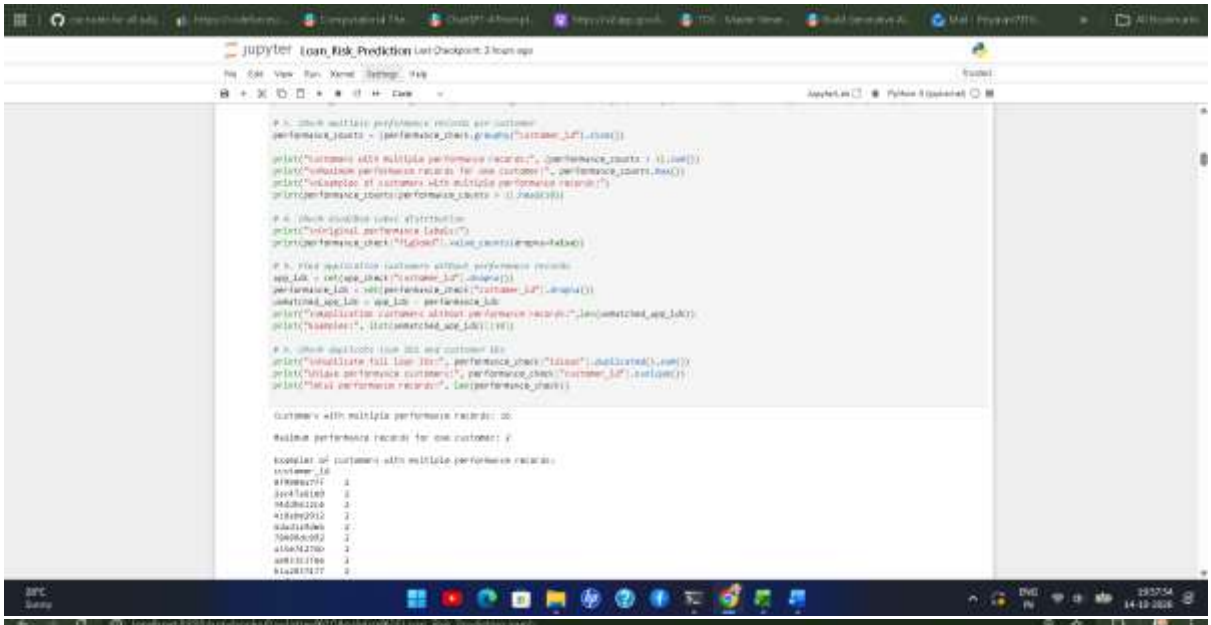
1. **Loan_Risk_Prediction.ipynb** — Jupyter Notebook containing the analysis, preprocessing, modeling, and evaluation.
2. **Loan_Risk_Prediction_Report.docx** — This technical report.
3. **loan_risk_predictions.csv** — Applicant-level predicted probabilities and risk categories.
4. **loan_risk_summary.csv** — Summary of applicant risk distribution and probability statistics.

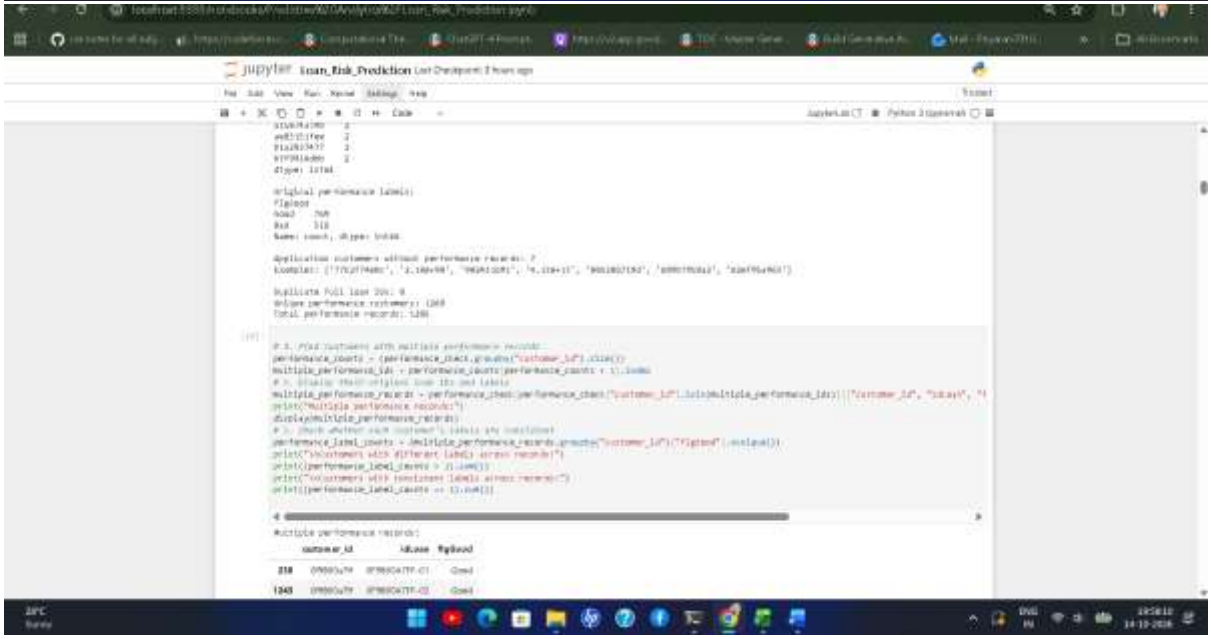
Additional model and configuration files can be supplied if required.

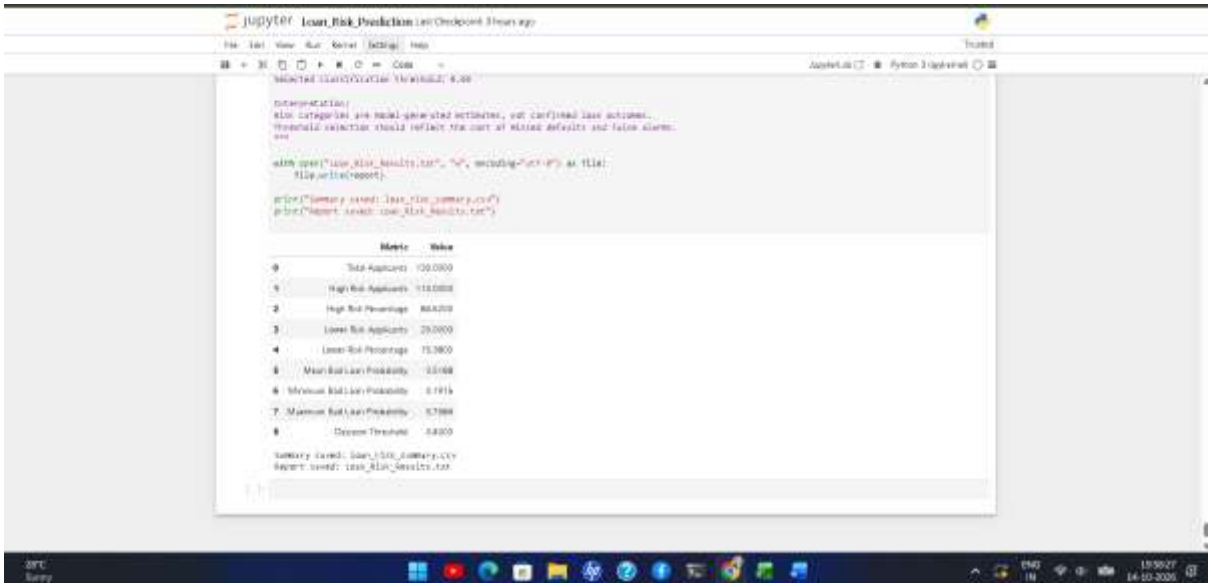


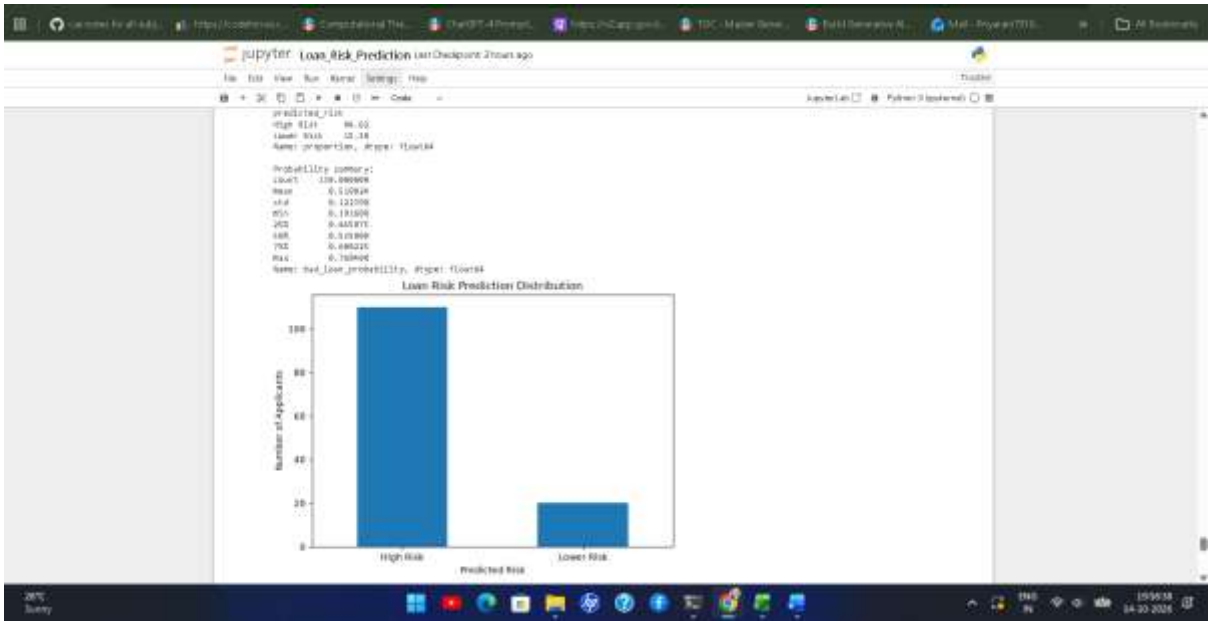



jupyter Loan_Risk_Prediction
Application customers: 647
Performance customers: 1269
Unique loan IDs: 1285
Duplicate application customer IDs: 0
Duplicate outcome loan IDs: 0
Matching customer IDs: 0
Application ID examples:
jupyter Loan_Risk_Prediction
Matching customer IDs: 669
Application ID sample length:
[10, 10, 10, 10, 10, 10, 10, 10, 10, 10]
Outcome ID sample length:
[10, 10, 10, 10, 10, 10, 10, 10, 10, 10]


jupyter Loan_Risk_Prediction Last Checkpoint: 3 hours ago
3. Check multiple performance records per customer
performance_counts = (performance_check.groupby("customer_id").size())
print("\nCustomers with multiple performance records:", (performance_counts > 1).sum())
print("\nMaximum performance records for one customer:", performance_counts.max())
print("\nExamples of customers with multiple performance records:")
print(performance_counts[performance_counts > 1].head(10))
4. Check disk0ue label distribution
print("\nOriginal performance labels:")
print(performance_check["fiplood"].value_counts(dropna=False))
5. Find application customers without performance records
app_ids = set(app_check["customer_id"].dropna())
performance_ids = set(performance_check["customer_id"].dropna())
unmatched_app_ids = app_ids - performance_ids
print("\nApplication customers without performance records:",len(unmatched_app_ids))
print("Examples:", list(unmatched_app_ids)[:10])
6. Check duplicate loan IDs and customer IDs
print("\nDuplicate full loan IDs:", performance_check["loaun"].duplicated().sum())
print("Unique performance customers:", performance_check["customer_id"].nunique())
print("Total performance records:", len(performance_check))
Customers with multiple performance records: 10
Maximum performance records for one customer: 2
Examples of customers with multiple performance records:
customer_id
0f90e6a77f 2
2ce47ab1d0 2
74d28e1216 2
418d9d2912 2
b3a31dfdeb 2
5b6964c952 2
a1de742780 2
a0631c1f66 2
61a2817477 2


jupyter Loan_Risk_Prediction Last Checkpoint: 3 hours ago
a1de742780 2
a0631c1fee 2
91a2817477 2
b7f91adeb0 2
dtype: int64
Original performance labels:
fiplood
Good 760
Bad 518
Name: count, dtype: int64
Application customers without performance records: 7
Examples: {'77f27f4e0c', '2.16e+08', '96b41c0f1', 'e.11e+11', '8651802f8d', '0d9979ba1c', '82ef95a46d'}
Duplicate full loan IDs: 0
Unique performance customers: 1268
Total performance records: 1286
3. Find customers with multiple performance records
performance_counts = (performance_check.groupby("customer_id").size())
multiple_performance_ids = performance_counts[performance_counts > 1].index
4. Display their original loan IDs and labels
multiple_performance_records = performance_check[performance_check["customer_id"].isin(multiple_performance_ids)][["customer_id", "loaun", "f
print("Multiple performance records:")
display(multiple_performance_records)
5. Check whether each customer's labels are consistent
performance_label_counts = (multiple_performance_records.groupby("customer_id")["fiplood"].nunique())
print("\nCustomers with different labels across records:")
print((performance_label_counts > 1).sum())
print("\nCustomers with consistent labels across records:")
print((performance_label_counts == 1).sum())
Multiple performance records:
customer_id loaun fiplood
218 0f90e6a77f 0f90e6a77f-01 Good
1245 0f90e6a77f 0f90e6a77f-02 Good


jupyter Loan_Risk_Prediction Last Checkpoint: 3 hours ago
Selected classification threshold: 0.60
Interpretation:
Risk categories are model-generated estimates, not confirmed loss outcomes.
Threshold selection should reflect the cost of missed defaults and false alarms.
"""
with open("loan_risk_results.txt", "w", encoding="utf-8") as file:
file.write(report)
print("Summary saved: loan_risk_summary.csv")
print("Report saved: loan_risk_results.txt")
Metric Value
0 Total Applicants 120.0000
1 High Risk Applicants 110.0000
2 High Risk Percentage 86.6200
3 Lower Risk Applicants 20.0000
4 Lower Risk Percentage 15.3800
5 Mean Bad Loan Probability 0.5168
6 Minimum Bad Loan Probability 0.1916
7 Maximum Bad Loan Probability 0.7388
8 Decision Threshold 0.6000
Summary saved: loan_risk_summary.csv
Report saved: loan_risk_results.txt


jupyter Loan_Risk_Prediction
Loan Risk Prediction Distribution
Number of Applicants
Predicted Risk
High Risk
Lower Risk